Most of a **galaxy's mass** is made up of **dark matter**.

It is estimated that there are **2 trillion galaxies** in the parts of **the universe** we can see.

Galaxies

Unimaginably huge collections of gas, dust, stars, and even planets, galaxies come in many shapes and sizes. Some are spirals, such as our own galaxy, others are like squashed balls, and some have no shape at all.

When you look up at the sky at night, every star you see is part of our galaxy, the Milky Way. This is part of what we call the Local Group, which contains about 50 galaxies. Beyond it are countless more galaxies that stretch out as far as telescopes can see. The smallest galaxies in the universe have a few million stars in them, while the largest have trillions. The Milky Way lies somewhere in the middle, with between 100 billion and 1 trillion stars in it. The force of gravity holds the stars in a galaxy together, and they travel slowly around the center. A supermassive black hole hides at the heart of most galaxies.

Astronomers have identified four types of galaxies: spiral, barred spiral, elliptical, and irregular. Spiral galaxies are flat spinning disks with a bulge in the center, while barred spiral galaxies have a longer, thinner line of stars at their center, which looks like a bar. Elliptical galaxies are an ellipsoid, or the shape of a squashed sphere—these are the largest galaxies. Then there are irregular galaxies, which have no regular shape.

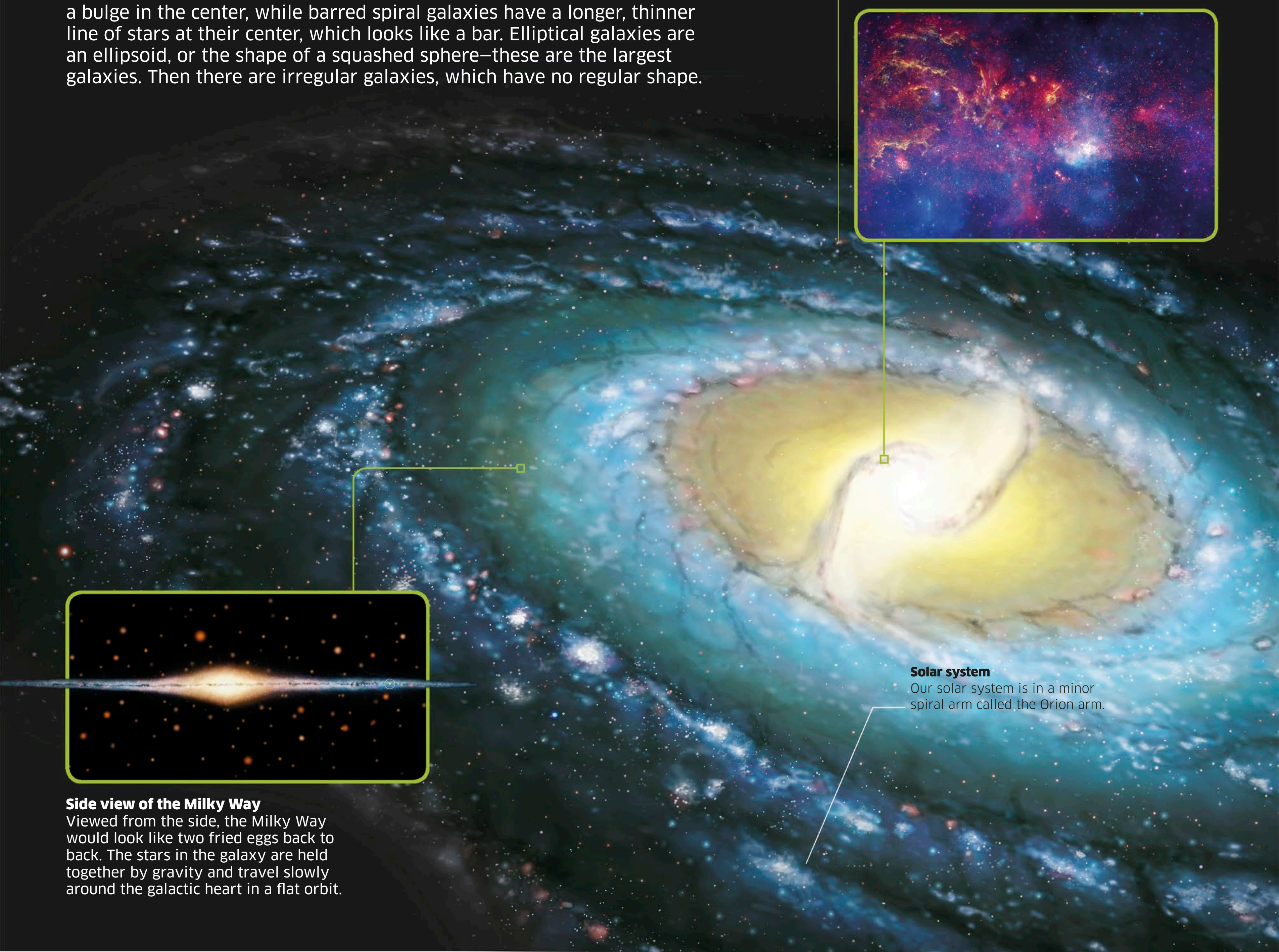
MILKY WAY

Type: *Barred spiral*

Diameter: *100,000 light years*

Our own galaxy, the Milky Way, is thought to be a barred spiral shape, but we cannot see its shape clearly from Earth because we are part of it. From our solar system, it appears as a pale streak in the sky with a central bulge of stars. From above, it would look like a giant whirlpool that takes 200 million years to rotate.

Galactic core
Infrared and X-ray images reveal intense activity near the galactic core. The galaxy's center is located within the bright white region. Hundreds of thousands of stars that cannot be seen in visible light swirl around it, heating dramatic clouds of gas and dust.



Solar system
Our solar system is in a minor spiral arm called the Orion arm.

Side view of the Milky Way
Viewed from the side, the Milky Way would look like two fried eggs back to back. The stars in the galaxy are held together by gravity and travel slowly around the galactic heart in a flat orbit.